

## Multiple Choice

1. **Answer:** A

*Options:* A) 0.4752; B) 0.6210; C) 0.4123; D) 0.5678; E) 0.5439

*Note:* Correct option: A - 0.4752

2. **Answer:** A

*Options:* A) The product of 7 x 5 is odd because both of the factors are even.; B) The product of 3 x 3 is odd because one of the factors is odd.; C) The product of 5 x 2 is even because both of the factors are even.; D) The product of 6 x 2 is odd because one of the factors is odd.; E) The product of 2 x 2 is odd because both of the factors are even.

*Note:* Correct option: A - The product of 7 x 5 is odd because both of the factors are even.

3. **Answer:** A

*Options:* A) 0.3524; B) 0.2123; C) 0.5948; D) 0.3187; E) 0.4298

*Note:* Correct option: A - 0.3524

4. **Answer:** A

*Options:* A) 25; B) 22; C) 30; D) 27; E) 29

*Note:* Correct option: A - 25

5. **Answer:** A

*Options:* A) 3.1415926; B) 4.6692016; C) 1.5707963; D) 0; E) 2.7182818

*Note:* Correct option: A - 3.1415926

## True / False

6. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

7. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

8. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: '0.25'.

9. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

10. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

## Long Form Response

11. **Answer:** `[asy] draw((-7,0)--(11,0),Arrows); draw((0,-5)--(0,11),Arrows); label("x",(12,0)); label("y",(-1,11)); draw(Circle((2,3),5)); dot((2,3)); dot((-1,-1)); label("(2,3)",(2,3),W); label("(-1,-1)",(-1,-1),W); draw((-1,-1)--(2,-1),dashed+red); draw((2,-1)--(2,3),dashed+blue); draw((2,3)--(5,3),dashed+red); draw((5,3)--(5,7),dashed+blue); dot((5,7)); label("(?,?)",(5,7),E); [/asy]` Refer to the above diagram. Since the opposite ends of a diameter are symmetric with respect to the center of the circle, we must travel the same distance and direction from  $(-1, -1)$  to  $(2, 3)$  as we do from  $(2, 3)$  to the other endpoint. To go from  $(-1, -1)$  to  $(2, 3)$  we run 3 (left dashed red line) and rise 4 (left dashed blue line), so our other endpoint has coordinates  $(2 + 3, 3 + 4) = (5, 7)$ .

*Note:* Students should show all steps in their mathematical solution.

12. **Answer:**  $\frac{12}{7}$ . Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

*Note:* Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.